

## 11 Syllables and Suprasegmentals

---

Our discussion of the articulatory and acoustic characteristics of speech sounds has until now focused on individual vowels and consonants. However, spoken language rarely consists of isolated speech sounds. Instead, vowels and consonants are combined to form larger units such as syllables, phrases, and sentences. These larger units can also be described in terms of their articulatory and acoustic properties. The examination of aspects of speech that extend beyond individual vowels and consonants is known as the study of suprasegmentals or prosody. The term prosody is sometimes used as a synonym only for “intonation.” However, as we discuss in this chapter, intonation is only one instance of a range of phenomena properly known as “prosody,” which we use as a synonym for “suprasegmentals,” intonation being part of it.

The principal suprasegmental features are stress (how “prominent” a syllable is), length (or quantity), tone ( $F_0$  level or movement in a syllable), and intonation ( $F_0$  movement during a phrase). These features are independent of those required for the description of segmental properties. While the term *suprasegmental* implies a difference between properties of individual segments and properties over longer stretches of speech, it is important to consider prosodic structure as an integral part of the speech signal.

Before any introduction of suprasegmentals, we need to discuss the notion of the syllable since suprasegmental features such as stress and tone make reference to it. While most people have an intuitive notion of what a syllable is, it is quite difficult to define a syllable using phonetic criteria.

### 11.1 Syllables

---

While syllables most often consist of a combination of consonants and vowels, they can also consist of individual vowels (*eye* [aɪ]) or consonants (the syllabic nasal in *cotton* [kɑtŋ]). The syllable is often represented by the Greek symbol σ. For descriptive purposes, a syllable can be said to consist of, minimally, a nucleus and, maximally, an onset, nucleus, and coda. Typically, the nucleus of the

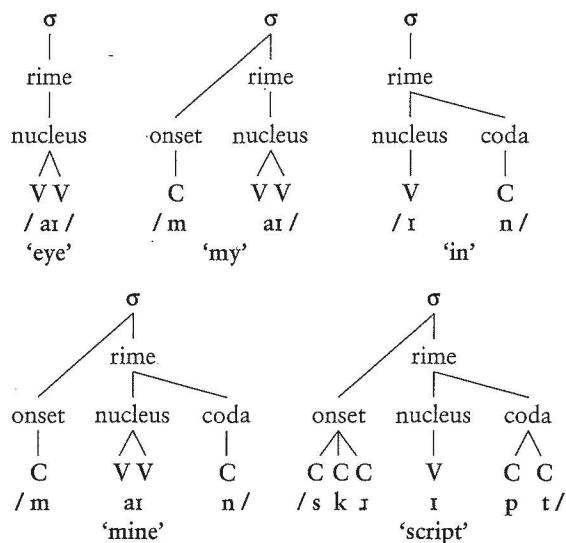


Figure 11.1 Different syllable types with their components ( $\sigma$  is a convenient symbol for "syllable").

syllable is the vowel, the onset is made up by the consonants preceding the vowel, and the coda consists of the consonants following the vowel. The nucleus and coda together form the rime (see Figure 11.1).

Most people can easily identify syllables. For example, native speakers of English generally agree that the word *yesterday* ['jes.tə.der] consists of three syllables (following IPA convention, syllable breaks are indicated by the symbol [.] ). However, other cases are more controversial. Words like *mirror* or *fire* will be perceived by some as having two syllables while others will hear only one syllable. A definition of the syllable should be able to account not only for the clear-cut cases but also for the "controversial" cases.

Phonetic definitions of the syllable have been proposed in both articulatory and auditory terms. Early articulatory accounts of the syllable claimed that each syllable was the result of a contraction of the respiratory muscles. However, subsequent work showed that there was no one-to-one mapping between syllables and muscle activity: certain bisyllabic (two-syllable) words were shown to involve only a single contraction of the expiratory muscles, and, conversely, certain monosyllabic (one-syllable) words were produced with two muscular contractions. (See Section 5.1.4 on the difficulty of relating muscular activity of the thorax to the loudness of speech sounds.)

Auditory definitions of the syllable make reference to the notion of *sonority*, the relative loudness of sounds. Even if all speech sounds were produced with the same length, stress, and pitch, some are louder than others. For example, vowels are louder than consonants, and within the class of vowels, [a] is louder than [i]. When speech sounds are ranked in terms of their loudness, a ranking

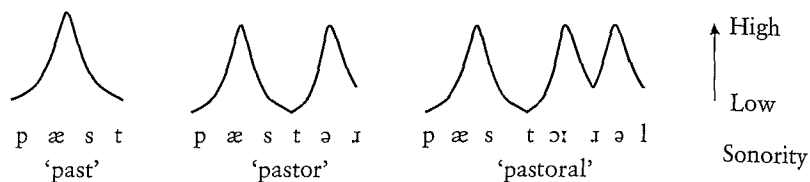


Figure 11.2 Idealized sonority levels of the words *past*, *pastor*, and *pastoral*.

known as sonority hierarchy is created. The two primary contributors to the perception of sonority are degree of opening of the vocal tract (the more open the tract, the louder the sound is) and voicing. Going from most to least sonorous, the order in this hierarchy is vowels → approximants → nasals → fricatives → plosives. Additional fine-grained distinctions can be made within each manner class, such as low vowels and voiced plosives being more sonorous than high vowels and voiceless plosives, respectively.

How does the sonority hierarchy help in defining the syllable? Phonologists have proposed that each syllable is associated with a peak in sonority (e.g. Selkirk 1984). The sonority principle stipulates that sonority increases from syllable onset, reaching a maximum at syllable nucleus, and then decreases toward syllable coda. Words such as *past*, *pastor*, and *pastoral* all fit the pattern predicted by the sonority principle, with one, two, and three sonority peaks, respectively (see Figure 11.2). Unfortunately, some English words violate the principle, most notably those beginning with [s]+plosive clusters such as *speak* [spik]. Since [s] is judged as more sonorous than all other voiceless and voiced plosives, a word such as *speak* would have two sonority peaks, yet it consists of only a single syllable. The sonority principle is perhaps best thought of as a statistical fact over many languages rather than a grammatical fact. There are violations in many languages that may have onsets like [mr,ft,rt] or codas like [rj]. Nevertheless, taking “sonority” as an ordering principle of an “idealized” prominence of sounds that roughly relates to the openness of the vocal tract and allowing for certain exceptions in a particular language gives a formal means of segmenting words into syllables.

In sum, while the notion of a syllable is intuitively straightforward and plays a prominent role in phonological theory, a comprehensive phonetic definition still remains to be formulated.

## 11.2 Stress

Stress is a property of a syllable that serves to make it relatively more prominent. Three levels of stress, from strongest to weakest, are usually distinguished: primary stress, secondary stress, and unstressed. For example, in the word *phonetician*

[fou.nə.'ti.n], the third syllable receives primary stress (indicated by a high vertical stroke ['] preceding the syllable), the first syllable receives secondary stress (indicated by a low vertical stroke [ˌ] preceding the syllable), and the second and last syllables are unstressed. (Articulatorily) a stressed syllable is usually produced with greater physical effort. This may be reflected in an increase in activity of the laryngeal muscles (particularly the cricothyroid muscle), an increase in subglottal air pressure, or an adjustment in spectral tilt; that is, increasing the amount of high-frequency energy in relation to the low-frequency energy (Sluijter et al. 1997; see Section 5.2.3). (Acoustically) the four main acoustic correlates of stress are fundamental frequency, duration, intensity, and formant frequency pattern. (The effect of the spectral tilt has not yet been intensively investigated and will be left out of the subsequent discussion.) Compared to an unstressed syllable, a stressed syllable in English typically has a higher fundamental frequency, greater duration, greater intensity, and a less reduced ("schwa-like") quality. Stressed syllables may be characterized by any one or more of these correlates. In some languages, stress can be indicated by means of a decrease in fundamental frequency (e.g. Bengali; cf. Hayes and Lahiri 1991) or much more complex patterns (e.g. Welsh; cf. Williams 1983).

The understanding of the acoustic correlates of English stress has benefited from a series of studies on the perception of bisyllabic minimal pairs such as *the record* ['ɛ.kərd] and *to record* [rə.'kɔrd]. In English, bisyllabic nouns are often stressed on the first syllable (sometimes referred to as a "trochaic" stress pattern) while bisyllabic verbs are stressed on the second syllable (an "iambic" stress pattern). The transcription of the minimal pair clearly indicates that there is a difference in phonetic quality: the stressed first syllable contains a full vowel [e] while the unstressed first syllable contains the reduced vowel [ə]. Figure 11.3 shows oscillograms, spectrograms, as well as intensity and pitch contours of the word *record* spoken as a noun and as a verb. In this example, all four acoustic correlates of stress are present. A comparison of the stressed syllables to the unstressed syllables shows an increased fundamental frequency, longer duration, increased intensity, and modified formant frequency pattern. With respect to the role of fundamental frequency, it is sometimes observed that rather than an increase per se in fundamental frequency, a change in any direction is a correlate of stress. That is, a stressed syllable could have either a higher or lower fundamental frequency than its unstressed counterpart. Perception experiments with synthetic speech, where the segmental content remained the same while each correlate of stress was systematically varied, suggest that fundamental frequency and duration may be stronger cues to stress than intensity. As for formant frequency, the second formant frequency (F2) of the vowel in the *first* syllable is lowered for the verb while F2 in the *second* syllable is lowered for the noun. Vowel reduction is pervasive in English. Many unstressed syllables are realized in English as [ə]. This is not necessarily the case in other languages. In Spanish, for example, the quality of the vowel /o/ does not change as a function of stress in minimal pairs such as *miro* ['mi.ro] ('I watch') and *miro* [mi.'ro] ('he watched').

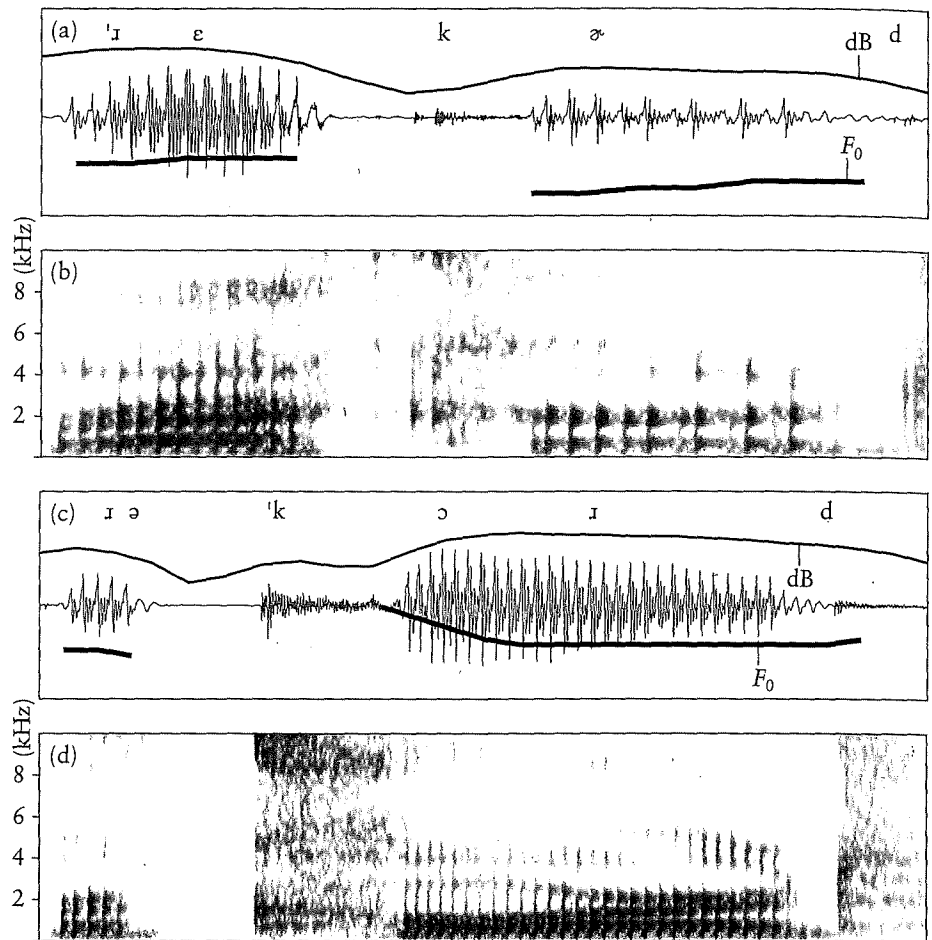


Figure 11.3 Oscillogram, spectrogram, intensity, and  $F_0$  contours of the word *record* spoken as (a, b) a noun ( $['rɛkərd]$ ) and (c, d) a verb ( $[rɪ'kɔːrd]$ ).

It must be kept in mind that the correlates of stress also serve other purposes. Fundamental frequency signals tone and intonation, duration is important in languages with phonemic distinctions in length, and formant frequency pattern is the primary correlate of vowel quality. The relative importance of a particular acoustic correlate of stress may in part be determined by its role in signaling other distinctions.

In addition to the syllable, the **foot** is also an important constituent in any discussion of stress. A foot can be defined as consisting of the stressed syllable and all unstressed syllables that follow it (for alternative definitions of the foot, see Hayes 1995). It is claimed that stressed syllables alternate with unstressed syllables to create a rhythm of prominence peaks. In a word such as *information*

[ɪn.fə.'meɪ.ʃn] the stressed syllables [ɪn] and [meɪ] are each followed by an unstressed syllable. Thus, the word *information* consists of two feet, [ɪn.fə] and [meɪ.ʃn]. Languages in which more and less prominent syllables alternate in this way are known as stress-timed languages. Within the group of stress-timed languages, a distinction is made between fixed and variable stress. In a language with fixed stress, primary stress always falls on the same syllable. For example, in Czech stress falls on the first syllable, in Polish words are almost always stressed on the penultimate syllable, and in French it is nearly always the final syllable. In variable stress-timed languages such as English or German, stress does not consistently fall on a fixed syllable in a word; it is variable across words (cf. Hayes 1995).

At the sentence level, the alternating stress creates different rhythms. For example, the utterance *Eddy's playing a trombone* consists of three feet, *Eddy's*, *playing a trom*, and *bone*. It is claimed that in stress-timed languages like English and other Germanic languages, linguistic rhythm is based on the isochrony of interstress intervals. That is, in stress-timed languages, interstress intervals tend to have the same duration. In other words, in the example utterance, each foot, regardless of the number of syllables (*Eddy's*, *playing a trom*, and *bone*), is equally long. However, as illustrated by the oscillogram in Figure 11.4, the even spacing of stress has received little empirical support.

In syllable-timed languages like French and Spanish, all syllables, stressed and unstressed, are assumed to occur at regular intervals, in contrast to stress-timed languages (in which stressed syllables are claimed to occur at regular intervals). Linguistic rhythm is thought to arise from the isochrony of syllables. In other words, for syllable-timed languages, stressed and unstressed syllables do not differ significantly in duration. Indeed, it has been reported that the difference between stressed and unstressed syllables is smaller in Spanish, a syllable-timed language, than in English, a stress-timed language. The fact that unstressed syllables are not reduced in Spanish also contributes to the regular intervals. However, syllable duration was not found to be constant.

We should note here that Japanese may provide an example of a third type of timing, mora timing. Japanese has a unit called the mora. In Japanese, the

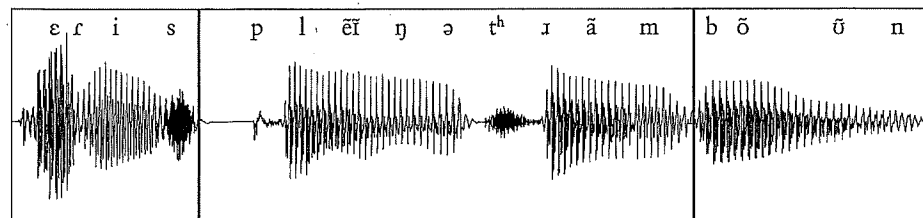


Figure 11.4 Oscillogram of the sentence *Eddy's playing a trombone*. The duration of each foot is marked and provides little support for the existence of isochrony.

difference between a syllable and a mora is that a consonant in the coda of a syllable forms a mora of its own (and so does the second half of a long vowel). According to traditional accounts, each mora takes up approximately the same duration. Consider the Japanese words /ha-fi/ 'edge' and /ke-k-ko-ŋ/ 'marriage.' While both words consist of two syllables, the first word has two morae and the second has four (separated by dashes for convenience). Some duration measures support a mora-based account. That is, /ke-k-ko-ŋ/ is twice as long as /ha-fi/. However, recent careful acoustic research has not been able to support the notion of the mora as a direct temporal unit (Warner and Arai 2001).

Depending on the structure of the sentence, a given word in a sentence can be pronounced with or without stress. In the sentence *I crave fame and fortune*, the word *and* receives emphasis and is produced in what is called its strong (unreduced) form, [ænd]. In contrast, in the sentence *I crave fame and fortune*, the word *and* is produced in its weak (reduced) form [ənd]. Function words (articles, conjunctions, prepositions, and pronouns) are usually much more prone to such strong-weak alternations than content words (such as nouns and verbs). Phonetically, this alternation can lead to changes in vowel quality, and even the loss of vowels or consonants, as in the weak forms of *and* [ənd], [ən], [nd], or [ŋ].

At the sentence level, the stress of a word can also change as a function of the stress of neighboring words. For example, the word *fourteen* produced in the utterance *I'll turn fourteen tomorrow*, is stressed on the second syllable. However, in the phrase *I am fourteen years old*, it is typically produced with stress on the first syllable. This phenomenon is known as stress shift or rhythm reversal. Note that *fourteen* is also stressed on the first syllable in the phrase *I am fourteen semesters behind*. The fact that *fourteen* is followed by an unstressed syllable demonstrates that stress shift is not triggered by a sequence of two stressed syllables. Instead, stress shift operates at the level of the phrase. As Carr (1999) points out, stress shift "reverses a sequence of a secondary stressed syllable and a primary stressed syllable when it is followed by a primary stressed syllable within the phrase" (p. 111).

When a word occurs as part of a sentence, a pitch accent or nuclear accent can be used to make certain stressed syllables more prominent than others. A pitch accent is any pitch configuration that serves to make a specific syllable in an utterance relatively more prominent. A syllable associated with a pitch accent is more stressed than any syllable that is not accented. In English and other stress-timed languages, the accented syllable is always a stressed syllable. The last accented syllable of an intonational phrase (see Section 11.4.2) is also more prominent than any other accented syllable and is therefore said to have a special type of pitch accent known as nuclear accent. It is this syllable that people typically refer to when they speak of 'the stressed syllable'. The location of the nuclear accent is determined by discourse properties that the speaker wants to highlight for the hearer (e.g., whether new information is brought into a conversation, or whether two items are contrasted).

### 11.3 Length

The acoustic correlate of length is duration. Duration is affected by many factors whose domain ranges from the segmental to the sentential level. Much of the current understanding of both local and global influences on segment duration stems from research in speech synthesis. Early attempts to synthesize speech (i.e. generate speech by computer) sounded incomprehensible at times because the synthesis rules did not adjust segment durations appropriately. This inspired researchers to conduct detailed studies to identify and quantify factors that affect duration. Table 11.1 lists some of these factors; it is based on a similar table by Klatt, one of the pioneers of modern speech synthesis.

Articulatorily, duration is determined by the timing of articulatory movements. The duration of the segment, syllable, foot, and sentence is affected by a number of both local and global factors. One factor that affects duration strongly is the presence of a phonemic length distinction, known as contrastive length. Many languages contrast long and short vowels. Duration seems to be the primary cue to vowel length in languages such as Arabic, Danish, Estonian, Ewe, Finnish, and Japanese, with long to short vowel duration ratios ranging from 1.3:1 to 2:1, that is, with long vowels being from 30 to 100 percent longer than their short counterparts. In these cases, the long vowel is indicated by the length mark [ː], as in [iː] versus [i]. In languages such as Dutch, English, and Swedish, however, the distinction is primarily one in terms of quality and only secondarily in terms of duration. In these cases, the contrast is represented by the use of two distinctive phonetic symbols, as in [i] versus [ɪ].

Although less common than vowel length contrasts, contrasts between long and short consonants also occur in a variety of languages. Italian is an oft-cited example with minimal pairs such as *fato* ['fato] 'fate' and *fatto* ['fatːo] 'done.' These long consonants are known as geminates. Other well-documented languages with a distinction between single and geminate plosives include Arabic, Bengali, Estonian, Finnish, Italian, Luganda, Swiss German, and Turkish, with long consonants generally being two to three times as long as their short counterparts. Geminate plosives can be realized by means of a longer closure duration (e.g. Swiss German) and/or a longer VOT (e.g. Cyprus Greek).

Intrinsic duration differences constitute another factor affecting length. When all other factors are held constant, low vowels are slightly longer (20–25 ms) than high vowels. This difference may be perceptible since psychophysical studies suggest that intrinsic duration differences exceed the minimal difference (just noticeable difference (JND)) needed by listeners to discriminate two sounds in terms of duration. Intrinsic duration differences have been reported for a variety of languages and are therefore likely to be language-universal. The most common explanation for the correlation between tongue height and duration is that low vowels involve a greater extent of articulatory movement. The tongue in a low

**Table 11.1** Parameters affecting the duration of segments (after Klatt 1973). Klatt gives references for every statement, which we have omitted here

|    |                                       |                                                                                                                                                                                                                                                                                                                                                        |
|----|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Pause insertion rule                  | Insert a brief pause before each sentence-internal main clause and at other boundaries limited by an orthographic comma.                                                                                                                                                                                                                               |
| 2  | Clause-final lengthening              | The vowel or syllabic consonant in the syllable just before a pause is lengthened. Any consonants in the rhyme (between this vowel and the pause) are also lengthened.                                                                                                                                                                                 |
| 3  | Phrase-final lengthening              | Syllabic segments (vowels and syllabic consonants) are lengthened if in a phrase-final syllable. Durational increases at the noun-phrase/verb-phrase boundary are more likely in a complex noun phrase or when subject-verb-object order is violated; durational changes are much less likely for pronouns. The lengthening is perceptually important. |
| 4  | Non word-final shortening             | Syllabic segments are shortened slightly if not in a word-final syllable. (This rule is disputed.)                                                                                                                                                                                                                                                     |
| 5  | Polysyllabic shortening               | Syllabic segments in a polysyllabic word are shortened slightly. (This rule is also disputed.)                                                                                                                                                                                                                                                         |
| 6  | Non-initial-consonant shortening      | Consonants in non-word-initial position are shortened.                                                                                                                                                                                                                                                                                                 |
| 7  | Unstressed shortening                 | Unstressed segments are shorter and more compressible than stressed segments.                                                                                                                                                                                                                                                                          |
| 8  | Lengthening for emphasis              | An emphasized vowel is significantly lengthened.                                                                                                                                                                                                                                                                                                       |
| 9  | Postvocalic context of vowels         | The influence of a postvocalic consonant (in the same word) on the duration of a vowel is such as to shorten the vowel if the consonant is voiceless. The effects are greatest at phrase and clause boundaries.                                                                                                                                        |
| 10 | Shortening in clusters                | Segments are shortened in consonant-consonant sequences (disregarding word boundaries, but not across phrase boundaries).                                                                                                                                                                                                                              |
| 11 | Lengthening due to plosive aspiration | A stressed vowel or sonorant preceded by a voiceless plosive is lengthened.                                                                                                                                                                                                                                                                            |

vowel has to move further to following consonants, which require a more closed vocal tract that usually goes along with a higher tongue position, and hence require more time. For consonants, it has been reported that labials tend to be longer than either alveolars or velars. One explanation is that the lips are less mobile (and hence, slower) articulators than the tongue.

There are additional local conditioning factors for segment duration. One such factor was discussed in Section 3.4.2.1, namely that vowels are longer when followed by a voiced as compared to a voiceless consonant. Other factors influencing segment duration include the place and manner of the consonants that either precede or follow the vowel. More global conditioning factors include the number of segments in the syllable and the number of syllables in the word. In general, segment duration is shortened as the number of segments and syllables increases. For example, the duration of the vowel [u] will be shortest in the word *fruitier*, somewhat longer in *fruity* and longest in *fruit*. This may again be an instance of isochrony, shortening longer words in an attempt to keep the intervals between stressed syllables relatively constant at the phrasal level.

In addition to local factors, global factors affecting duration include the position of the segment or syllable in the sentence. This effect can be observed by measuring the length of a syllable in running speech and comparing it to the same syllable in a different position. A syllable is substantially longer at the end of a phrase than at the beginning or middle of a phrase, a phenomenon known as phrase-final lengthening.

One final global factor affecting segment length is speaking rate. Speaking rate can be defined as the overall tempo of a speaking turn. The same sentence (*How do you do?*) can be uttered at a slow or a fast tempo. Speaking rate can be computed based on the number of phonemes or syllables per second. For English, the average speaking rate is on the order of 4-6 syllables per second. However, with an increase in speaking rate, there is often a change in the phonemes that are produced. It makes a difference, then, whether rate is computed on the basis of the idealized transcription of an utterance or on the actual produced sounds.

When calculating speaking rate, filled pauses (e.g. "um," "oh," etc.) within utterances as well as silent pauses between the utterances that make up the speaking turn must also be taken into account. Increases or decreases in speaking rate shorten or lengthen, respectively, both segment durations and pauses. However, changes in speaking rate do not affect all segment durations and pauses equally. In general, longer segments are affected more than shorter ones so that the duration of vowels, for example, changes more than that of consonants. In addition to affecting duration, variations in speaking rate can also affect  $F_0$  range, with an increase in speaking rate typically leading to a reduction in  $F_0$  range.

## 11.4 Tone and Intonation

While a change in  $F_0$  serves as a cue to lexical stress, a number of local and global factors affect  $F_0$  per se. As mentioned in Section 7.3.1.4, pitch is the perceptual correlate of fundamental frequency, which makes vocal fold vibration its articulatory correlate. The most obvious factor affecting fundamental frequency is gender, with average fundamental frequency values of 130 Hz for males and 220 Hz for females (although these values may vary across cultures). Another factor affecting fundamental frequency is the speaker's emotional state. For example, a substantially raised  $F_0$  seems to be indicative of anger, fear, or happiness. Vowels also differ in terms of their intrinsic fundamental frequency.<sup>1</sup> In general, the fundamental frequency of high vowels is higher (by approximately 15 Hz) than that of low vowels. This is well within a listener's range of sensitivity since the just noticeable difference for fundamental frequency is approximately 1–2 Hz under ideal conditions and 10 Hz with speech-like stimuli. (Klatt 1973).

Intrinsic fundamental frequency differences have been observed across the world's languages. One explanation states that the effect is due to the muscular link between larynx and tongue. In this "tongue-pull" theory (Lehiste 1970), raising the tongue for the production of high vowels causes the larynx to rise as well, which leads to a tilt between cricoid and thyroid and increases the tension of the vocal folds, ultimately resulting in an increase in fundamental frequency. A general physiological "laryngeal dependency" theory has also been proposed (Ladd and Silverman 1984), according to which greater muscular tension for higher vowels leads to a greater muscular tension in the larynx. Yet another approach is "acoustic coupling" (Flanagan and Landgraf 1968), according to which the narrowed vocal tract leads to a Bernoulli effect with a higher speed of airflow, which reduces the air pressure above the larynx, which increases trans-glottal pressure, which ultimately increases the oscillation frequency of the vocal folds. Finally, the  $F_0$  could be under "active speaker control" to enhance vowel height contrasts (Kingston & Diehl 1994).

Similar to local conditioning factors for duration, a segmental conditioning factor for  $F_0$  is the observation that a vowel's fundamental frequency is slightly higher (approximately 5–10 Hz) when the vowel is preceded by a voiceless obstruent as compared to a voiced obstruent.

### 11.4.1 Tone

While gender is a major contributor to variation in fundamental frequency across speakers, the factor that affects fundamental frequency most strongly for a given speaker is the presence of a phonemic fundamental frequency distinction, known as contrastive tone or lexical tone. Tones can be defined as pitch variations that

change either the lexical or grammatical meaning of a word. A language in which the meaning of a word depends on its tone is known as a **tone language**.

Tone languages can have either **register tones** or **contour tones**. In a register tone language, the relative height of each syllable's pitch within the speaker's pitch range (or register) provides a cue to word meaning. These languages typically have up to four distinctive pitch levels. The vast majority of languages spoken in Africa are register tone languages (including Igbo, Shona, Yoruba, and Zulu to name but a few). Yoruba, for example, has three register tones, high, as in the first and second syllable of [ó.bé] 'he jumped,' mid, as in the second syllable of [ó.bē] 'he is forward,' and low as in [ó.bè] 'he asks for pardon.' Notice that, following IPA convention, the high tone is marked by an acute accent ['] over the vowel, the mid tone by a macron [¯], and the low tone by a grave accent [˘].

In a **contour tone language**, pitch movement, instead of pitch level, serves to distinguish word meaning. Each tone is associated with a particular shape and direction of the pitch trajectory. Many of the languages spoken in Southeast Asia (including Mandarin Chinese, Thai, and Vietnamese) are contour tone languages. Mandarin Chinese, for example, has four tones, high-level (Tone 1), high-rising (Tone 2), low-falling-rising (Tone 3), and high-falling (Tone 4). Figure 11.5 shows pitch contours for each of the four tones. Tone 1 is indeed at the high end of the speaker's range and remains level throughout its duration. Tone 2 starts lower but still in the speaker's upper range and rises throughout its duration. Tone 3 starts toward the speaker's lower range, dips even lower, and then rises toward the end. And Tone 4 starts about as high as Tone 1 but then shows a rather steep fall.

Table 11.2 shows sample transcriptions of the four tones. The column labeled "Pitch" indicates a common numerical representation of the tones, where 1 and 5 specify the lowest and highest points, respectively, of a speaker's pitch range, and 2, 3, and 4 are equally spaced in between (that is, "35" represents a tonal contour that starts in the middle and rises to the highest pitch of a speaker's range). IPA has two alternative sets of symbols to indicate tone, which we label as "tone markers" and "tone letters." Both sets are listed under "tones and word accents" on the IPA chart. The tone markers are placed above the vowel. In Table 11.2, the column labeled "Tone markers" shows the four markers used to indicate the Mandarin tones. Tone letters follow the syllable and consist of a vertical line with a stroke preceding it. The vertical line represents pitch range, and the stroke indicates movement and height within that range, as shown in the column labeled "Tone letters."

Historically, the process by which a non-tonal language changes into a tonal language is known as tonogenesis. We will focus on just one of the ways in which lexical tones may develop, having to do with a segmental conditioning factor mentioned previously, namely the fact that the fundamental frequency is higher for a vowel preceded by a voiceless as compared to a voiced obstruent. Research suggests (cf. Hombert 1978) that some tonal distinctions arose as the result of the listener's reinterpretation of a previously secondary cue after the disappearance of the primary cue. For example, in a language with voiced and

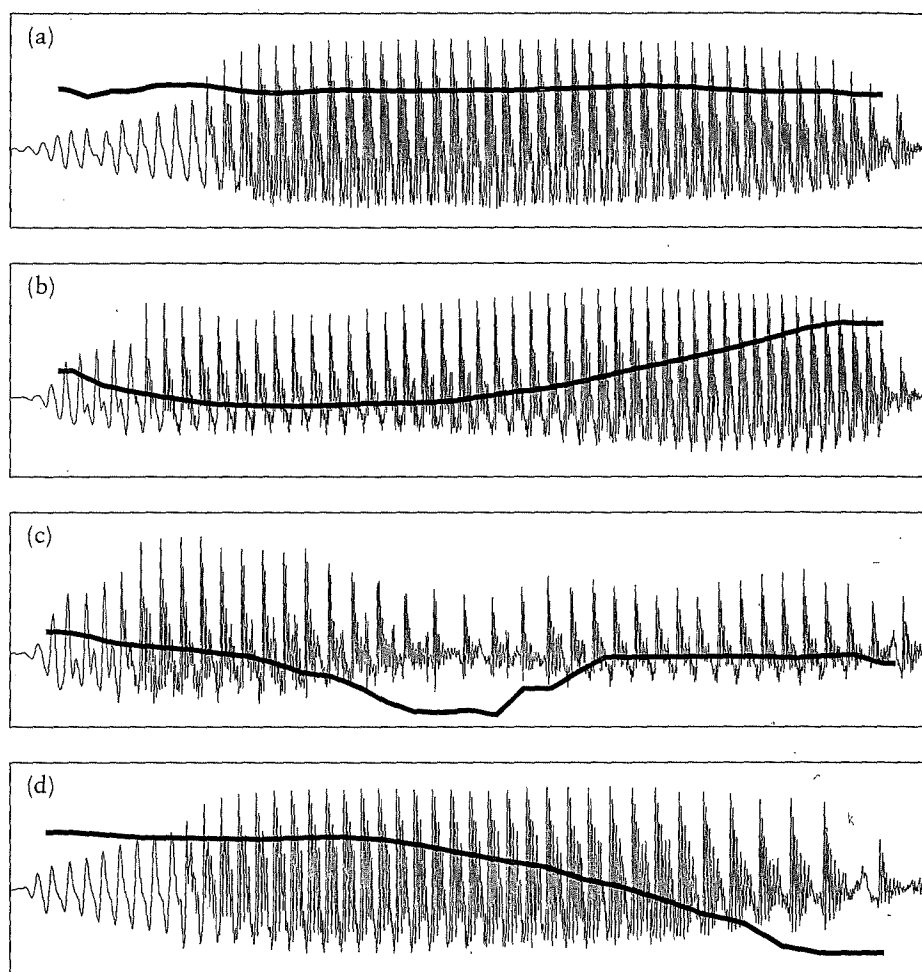


Figure 11.5 Oscillograms and pitch contours (dark lines) of the word [ma] spoken with each of the four Mandarin tones by an adult male.

Table 11.2 The four tones of Mandarin Chinese and three ways of transcribing them

|                            | <i>Pitch</i> | <i>Tone<br/>markers</i> | <i>Tone<br/>letters</i> | <i>Gloss</i> |
|----------------------------|--------------|-------------------------|-------------------------|--------------|
| Tone 1: high-level         | 55           | [mā]                    | [ma ˥]                  | 'mother'     |
| Tone 2: high-rising        | 35           | [má]                    | [ma ˨˥]                 | 'hemp'       |
| Tone 3: low-falling-rising | 214          | [mǎ]                    | [ma ˨˥˨]                | 'horse'      |
| Tone 4: high-falling       | 51           | [mà]                    | [ma ˨˥˨˥]               | 'scold'      |

voiceless obstruents, the vowels following voiced obstruents have a slightly lower pitch. As mentioned earlier, this may be actively produced by the speakers of a language to enhance the voicing contrast. However, when the language subsequently loses its voicing contrast in the plosives, the originally secondary pitch difference now becomes the primary cue to the lexical distinction and a tone language with two tones, high and low, on the vowels, has emerged. The tone contrast, which was originally only a by-product of the voiced-voiceless distinction, has then become part of the phonemic inventory of a language.

### 11.4.2 Intonation

Intonation is the distinctive use of pitch over units larger than a single word. Intonation may convey linguistic information by marking the boundaries of syntactic units or by distinguishing different sentence types such as statements, questions, and commands. Intonation may also convey non-linguistic information such as boredom, impatience, or politeness. We will limit our discussion to the linguistic functions of intonation.

Many languages indicate the end of a syntactic unit such as a phrase or sentence by means of a falling pitch (such as in declarative sentences). Many languages mark a yes-no question (a question to which the answer is expected to be either "yes" or "no") or other requests for information with a rising intonation. Although English declarative sentences and questions usually differ in terms of word order (*This is Jane.* vs. *Is this Jane?*), the same declarative sentence could be a statement or question, depending on its intonation. For example, the statement *We've been invited to a party tomorrow.* with a falling intonation informs somebody about an invitation whereas *We've been invited to a party tomorrow?* with a rising intonation expresses surprise about the existence of such an invitation.

Pitch contours for a declarative statement and a yes-no question are shown in Figure 11.6. Notice that the declarative sentence is marked by a general lowering of the fundamental frequency throughout the sentence, a phenomenon known as declination. In contrast, for the yes-no question a steep rise is evident, particularly at the end of the sentence.

The declination that is often observed in English declaratives can be perturbed by local changes in emphasis. Consider the following sentences, in which the emphasized word is in italics:

- a We will buy our *first* home tomorrow.
- b We will buy our first *home* tomorrow.
- c We will buy our first home *tomorrow*.

Pitch contours for each of these three readings are shown in Figure 11.7. The intonation contours show a clear change in pitch at the tonic syllable, the syllable of the phrase receiving primary stress (the italicized syllable in the example sentences).

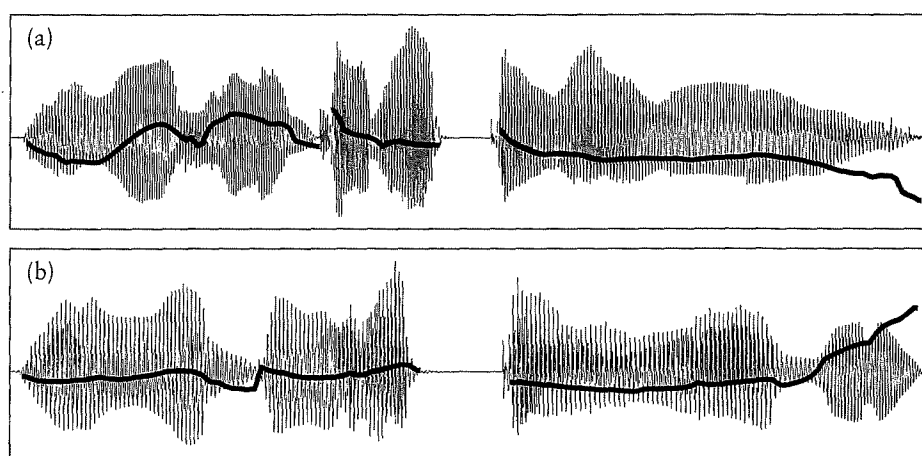


Figure 11.6 Oscillograms and pitch contours for *We've been invited to a party tomorrow* as (a) a declarative sentence, and (b) a yes-no question.

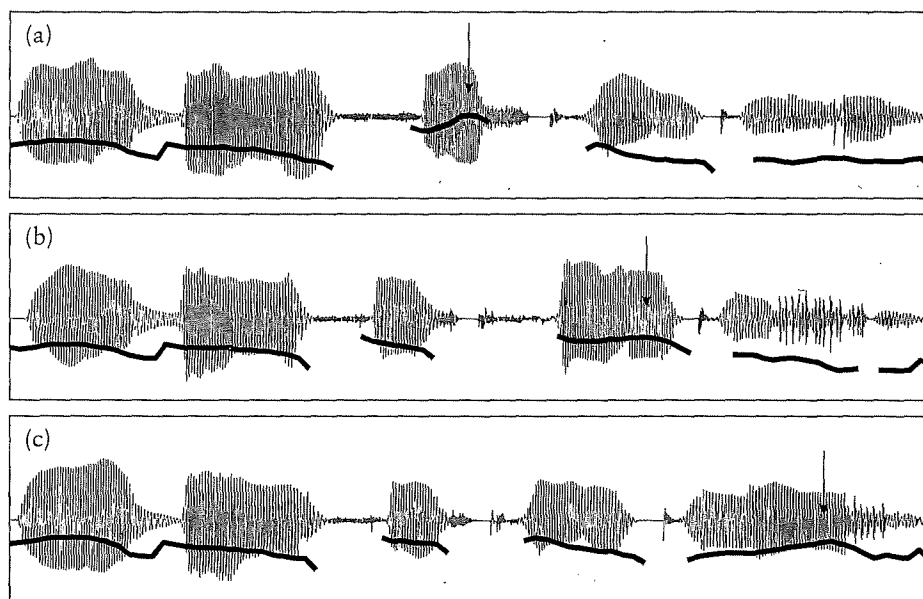


Figure 11.7 Oscillograms and pitch contours of the sentence *We will buy our first home tomorrow* with emphasis on (a) "first," (b) "home," and (c) "tomorrow" as indicated by the arrows.

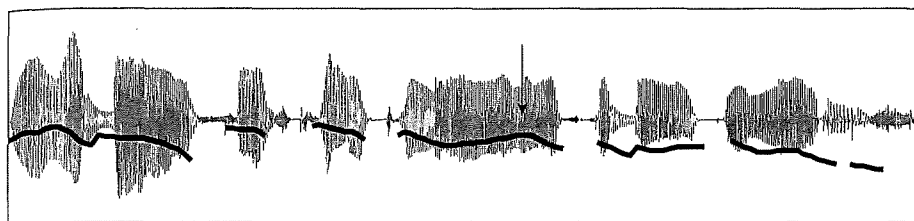


Figure 11.8 Oscillograms and pitch contour of the sentence *We will buy our first home tomorrow if the bank will let us*, consisting of two intonational phrases. The arrow marks the onset of the second intonational phrase *if the bank will let us*.

All sentences that we have considered so far consist of a single intonational phrase. However, a sentence such as *We will buy our first home tomorrow if the bank will let us* consists of two intonational phrases. As shown in Figure 11.8, the onset of the second intonational phrase *if the bank will let us* is marked by an increase in fundamental frequency, which indicates the existence of a second intonational phrase.

Figure 11.9 shows the pitch contours of sentences that can be considered minimal pairs in terms of their syntactic structure. The sentence *Almost everyone knows but Linda hasn't found out yet* consists of two conjoined main clauses while the sentence *Almost everyone knows Belinda hasn't found out yet* is nearly identical in terms of its segments but consists of a main clause and embedded clause. The

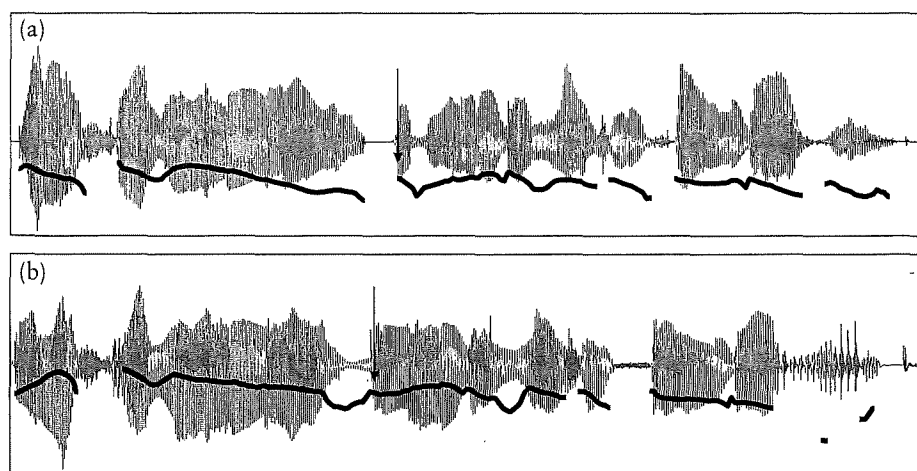


Figure 11.9 Pitch contours of the sentences (a) *Almost everyone knows but Linda hasn't found out yet* and (b) *Almost everyone knows Belinda hasn't found out yet*. The different syntactic structures of the two sentences are reflected in their distinct intonation patterns at the beginning of [lɪn], marked by the arrows.

pitch contours of these sentences are distinct in that the boundary between the two main clauses is marked by a more pronounced “fall-rise” pattern than that between main and embedded clause. While the peak value of  $F_0$  in *knows* was the same for both sentences, the lowest  $F_0$  value in *knows* and the peak value in *lin* were significantly lower (by approximately 5 Hz) for the sentence consisting of two conjoined clauses (Cooper and Sorensen 1977). Research also suggests that listeners can make use of this kind of information while parsing the sentence.

We have discussed stress, tone, and intonation as if they were all independent phenomena. However, as soon as the domain of investigation is larger than a single word, cues to stress, tone, and intonation begin to interact, as mentioned in our discussion of the role that intonation can play in making certain words more prominent. In addition, we do not mean to suggest that a language can use pitch for either tone or intonation only, or that a language can only be either a stress or a tone language (see Hyman 2006). In most tone languages, declarative statements are produced with a falling intonation (i.e. they use intonation). Moreover, several languages, including Pirahaã, Ma'ya, and Curaçao Papiamentu have been documented as having both stress and tone.

Finally, we end this chapter by discussing a language type that is not quite a tone language nor a stress language, again emphasizing the point that the distinctions and categories introduced in this chapter are not always clear-cut. This language type is the pitch-accent language, of which Japanese is a well-known example. Pitch-accent languages are similar to tone languages in that each syllable (or mora, in the case of Japanese) in a word is associated with a particular tone. However, unlike in a tone language, in a pitch-accent language, the entire tonal pattern of a word is known once the location of the accent of the word is known. Consider the examples from Japanese shown in Table 11.3.

The location of the accent, indicated by the diacritic ['], is not predictable and needs to be learned for each word. However, once the location of the accent is known, the entire tonal pattern follows from two simple rules. The first rule is that an accented mora and all morae preceding it receive a high tone (H) while all morae following the accented mora receive a low tone (L). The second rule is that (in the dialect spoken in Tokyo) the first mora of a word receives a low tone

**Table 11.3** Words illustrating Japanese (Tokyo dialect) as a pitch-accent language

| <i>Tonal pattern</i> | <i>Word</i> | <i>Gloss</i> |
|----------------------|-------------|--------------|
| H-L                  | [sóʔa]      | ‘sky’        |
| L-H                  | [kawá]      | ‘river’      |
| L-H-L                | [kokóʔo]    | ‘heart’      |
| L-H-H                | [otokó]     | ‘man’        |

when it is not accented. Once the location of the accent is known, these two rules account for all forms in Table 11.3. This is in contrast with tone languages such as Mandarin Chinese, where speakers must learn the pitch pattern of each syllable of a word.

In this chapter, we have discussed the principal suprasegmental features of stress, length, tone, and intonation and their acoustic correlates of fundamental frequency, duration, amplitude, and formant frequency patterns. Perceptually, the four dimensions available to the auditory system for differentiating aspects of the speech signal are pitch, duration, loudness, and auditory quality (= timbre). In the next chapter, we will discuss the basic structure and function of the human auditory system as well as some of the non-linear ways in which the auditory system transforms the speech signal.

## Exercises

---

- 1 What is prosody? What features are used to describe prosody?
- 2 Why is it difficult to define a syllable? What is the sonority hierarchy and how does it help?
- 3 What is stress and what are its acoustic correlates?
- 4 Briefly explain the difference between stress-timed, syllable-timed, and mora-timed languages.
- 5 How do register tone and contour tone languages differ?
- 6 What is the difference between intonation and stress?

## Note

---

- 1 Sometimes called "intrinsic pitch," although the term intrinsic fundamental frequency is to be preferred since the rate of vocal fold vibration is increased (and it is not a perception phenomenon) and the increased  $F_0$  does not necessarily result in an increase in perceived pitch.